Entry into the Stockholm Junior Water Prize 2023

Converting Water into Clean Home Energy

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2. Preliminary Matters

I. Abstract

We aimed to create a turbine that could harness the water that flows through the pipes in our homes everyday. We designed a turbine that was made to fit in household pipes and then tested the electricity production using multimeters and alternators we fabricated. Then, we tested the RPM of our turbine designs and recorded our findings. Additionally we used physics calculations to find theoretical yields and efficiency for our turbines. We found that our turbine systems all produced upwards of 100 RPM in their assemblies and the final prototype we created consistently produced greater than 220 RPM. Our assemblies produced constant high RPM and demonstrated the power within our walls. In order to change our prototype in the future we could implement it in a vertical pipe system in order to maximize the forces acting on the rotor and use gravity as well as the force of the water in the pipes. The stator could also be modified with the coil cores changed to metal and the stator expanded to make space for greater coil size.

II. Table of Contents

III. Key Words

Sustainability · Kinetic Energy · Tidal Power · Straflo Turbine · Impulse Turbine · Fluid Dynamics · Stator · Alternator · Neodymium Magnet · Hydroelectricity · 3D Printing · Water Pressure

IV. Abbreviations and Acronyms

CAD (Computer Aided Design) · TSA (Technology Student Association) · PLA (Polylactic acid) · SLS (Selective Laser Sintering) · FDM (Fused Deposition Modeling) · SLA (Stereolithography) · ABS (Acrylonitrile Butadiene Styrene) · PVC (Polyvinyl chloride) · UV (Ultra-violet) · EPA (Environmental Protection Agency) · RPM (revolutions per minute) · EIA (Energy Information Administration)

V. Acknowledgements

We would like to thank our teacher sponsor, Ms. Nunnink, who has supported us through this project by facilitating our ideas and always pushing us to do more. We would also like to thank our school and regional fair coordinators Mr. Sanderson and Mr. Hill. Additionally, we would like to thank our parents for helping us and supporting us throughout our project development. Next, we would like to credit John Franklin Crenshaw (PhD candidate at Washington U at St. Louis), Professor Luciano Castillo (professor of mechanical engineering at Purdue University), and Diego Andres Siguenza Alvarado (PhD Candidate at Purdue University) with their assistance in understanding fluid dynamics and other physics-related concepts pertaining to this project. Finally, we would like to thank Brady Scheinfeld for assisting us in our project development up until this stage when he was no longer able to assist because of prior commitments.

VI. Biography

Charles Fucile is an eleventh grade student at Midtown High School in Atlanta, Georgia in the United States of America. He swims for his summer league team and his school team as well as playing water polo on a club and school team. Charles participates in his school's speech team in Original Oratory where he delivered a speech about rejecting new ideas in the 2022-2023 school year. He won first place at the 1st/2nd Year State Tournament in speech and 4th place in the Varsity State Tournament. He also received second place in the TSA Extemporaneous Speech competition at the State Leadership Conference. Charles also won 3rd Place in 2021 in the Geospatial Technology competition at the TSA State Leadership Conference. He enjoys studying math, arts, science, and engineering. Charles also likes to spend his time playing video games and reading. Charles wants to study Aerospace or Mechanical Engineering.

Samuel Neff is also an eleventh grade student at Midtown High School in Atlanta, Georgia in the United States of America. He enjoys math, engineering, history and business. He is a captain on the varsity swim team and plays club and travel water polo. He spent his last two summers lifeguarding, going to water polo tournaments and has qualified to attend the Water Polo Junior Olympics both years with his travel team. Samuel was awarded his Eagle Scout in August of 2022 and makes his way back to support his troop. He is a part of the Future Business Leaders of America at his school and has attended their events over the past two years. In his spare time, he enjoys listening to music, his favorite artist is Kendrick Lamar, a large inspiration in Samuel's Life. Samuel intends to study Mechanical engineering with an automotive focus.

Last year Samuel and Charles qualified for the Georgia Science and Engineering Fair for their water polo project, they were also awarded 1st place at their regional competition. Samuel and Charles were inspired to pursue this project and this competition after hearing about the Stockholm Junior Water Prize at last year's state science and engineering fair.

3. Introduction

In 2020 the United States spent $1 trillion dollars on energy, and annual costs were approximately $3,039 per person [1]. Energy becomes a massive expense for many families across the United States because energy companies require monthly payments to maintain energy usage. However, systems of renewable energy in homes help to limit these expenses by making households self-sufficient and creating one-time expenses as opposed to recurring payments. Energy not only remains a financial problem, but also poses massive environmental danger as well. The energy sector is the largest emitter of greenhouse gasses, second only to the burning of fossil fuels for energy, heat, and transportation [2]. According to the United States EIA [3], water heating accounts for 19% of energy consumption in homes across the United States. By combining the energy production within the home with this system we can help contain these energy costs to the water system itself and attempt to make it entirely sustainable and self-supported. Not only can this relieve an economic strain, but this can also help to ease an environmental strain. The EPA [4] directly links greenhouse gasses to climate change which pose a massive threat to our environments and ecosystems. In order to reduce this damage and continual deterioration, clean energy is always being expanded upon. The natural forces around us are being harnessed in a variety of ways from solar to wind energy, but these generators are expensive and require lots of space to be effective. In order to reduce this massive economic weight and limit the ecological threat posed by climate change, we explored a smaller scale and potentially cheaper alternative to current sustainable energy production: small scale water turbines. According to the EPA [5] The average American family uses more than 300 gallons of water per day at home. While these turbines would not actively reduce the amount of water being used in households, these turbines would effectively reduce the amount of wasted water by making all the water in these pipes useful in the production of energy. The water flowing through our pipes, whether in our homes or businesses is a resource with untapped potential. We are currently letting the energy flowing through our walls pass by. Not only do American families use 300 gallons of water a day, but we are the nation with the second highest annual water usage of 817 trillion liters of water a year [6]. Texas and California lead this water usage from large irrigation efforts and farming. Our turbines could be implemented across all environments where water is flowing, from irrigated fields to industrial factories. This water is continuing to create droughts and climate change around the world as more and more water is pulled from our environment for minimal usage. Wastewater is a massive problem as much of the water that flows through our pipes is not even used. According to the EPA [7], Letting your faucet run for five minutes while washing dishes can waste 10 gallons of water and use enough energy to power a 60-watt light bulb for 18 hours. This demonstrates how energy and water are being wasted everyday. Water overuse continues to plague us as we allow water to be wasted by leaving faucets running or overwatering our lawns. While steps can also be taken to reduce our water usage and turn off faucets when we are not using them, implementing additional turbines helps to give new purpose to otherwise wasted water. While everyone should also take steps to reduce their water usage, our prototype takes advantage of the water already flowing through our pipes and maximizes the benefits that it provides. When all the water in our homes is being used to produce electricity it all becomes useful and the potential of the water flowing through our pipes is maximized. While there has not been any direct research or experimentation in the area of in-home hydroelectric power production, there are currently a plethora of hydroelectric rotor designs and systems. Hydroelectric generation continues to innovate and change, but there is a focus on larger scale hydroelectricity in environments like the ocean or dams. Turbines such as the Straflo Turbine design have been used in an array of applications such as producing tidal energy on the ocean floor. This specific usage of turbines influenced our choice to attempt to design a Straflo Turbine Assembly.

4. Materials and Methods

4a. First Experiment

Models were designed to fit a 3” diameter pipe with an external alternator box. The rotor (figure 5) spun an axle in a pipe (figure 6) that was connected to a chain belt leading to an alternator drum, connected to the alternator housing. This version of the project was designed in Fusion 360, a CAD software. The rotor, pipe, and alternator components were printed on an Ender 3 using PLA. This 3D printer uses a nose tip that liquefies the PLA and it cools upon contact. The pipe was fused to two 6” lengths of clear PVC pipe using PVC glue. This alternator featured 12 small coils with steel nut cores and a 6 magnet drum in the middle connected to the chain belt. The alternator did not function so we changed the design to eliminate possible problems with the design. We created an alternator with a greater proximity between the coils and magnets and removed the material that had originally been between them. This new alternator functioned to produce small amounts of power, but was largely inefficient. This alternator also did not connect to our rotor as the design had to be changed to accommodate the problems we had encountered. We placed the rotor in the pipe with the attached lego axle and gear. We decided to use a hose to water through the system and measured the RPM by attaching a string to one of the blades and letting the rotor spin for 10 seconds. Then, we multiplied this number by six to find the estimated RPM of the rotor over the course of a minute. We did this experiment six times to find the anticipated RPM of the system.

4b. Second Experiment

Our second experiment featured a complete redesign of our alternator system. We introduced the concept of a stator (figure 7), a ring of copper coils on the outside of the pipe (figure 8) and attached the magnets to the rotor (figure 9). We believed that this new version of the system would not only streamline maintenance, but the design, and efficiency as a whole. No longer would we have to rely on an external alternator, a chain, or any of the other moving parts previously exhibited in our first experiment. Now, the rotor was the only moving part containing just the magnets. This version was again designed on Fusion 360, but this time printed on a Fuse 1. This 3D printer prints with nylon 12 powder that was laser hardened. Following a failed print the redesigned pipe featuring an indentation to hold the stator it was reprinted on the Snapmaker 2.0 using PLA. This 3D printer prints the same as the Ender 3. We repeated the steps from the first experiment testing the RPMs in 10 second intervals, multiplying them by six, then repeating the trials 6 times. However, this time without the lego axle or gears, we introduced a ball bearing to the system providing more revolutions and efficiency to the system.

4c. Third Experiment

The third experiment was the same as experiment two, however, this revision scaled the project down from 3" diameter to 1.5" diameter. We did this to move from the size of drainage and ventilation pipes, to the standard household water pipe size. The pipe (figure 10) was made longer to reduce the number of parts, the rotors (figure 11) length was also increased to fit different magnets, no changes were made to the stator (figure 12) besides its ½ scale adjustment. These parts were printed on an Ender 3 using ABS. While made of a different compound, ABS and PLA share many properties except heat tolerances. We again, repeated the steps from the first experiment testing the RPMs in 10 second intervals, multiplying them by six, then repeating the trials 6 times. Now, with a smaller lighter system, we believed we could achieve higher RPM.

4d. Material Lists

Experiment one: PLA printed 3D parts (rotor, pipe, alternator housing pieces, and alternator drum), 12 3/8x1/2" hex bolts, 20 gauge copper wire, PVC glue, electrical tape, hot glue, sand paper, Dremel tool with sand paper tips, 2 3"x1' clear PVC pipes, two LEGO Technic 8 tooth gear (3647), 100 LEGO Technic link chain (3711), and one Lego Technic axle 5L (32073).

Experiment two: Nylon 12 3D printed parts (rotor and stator), PLA 3D printed part (pipe) 20 Gauge copper wire, a wooden dowel, 6 1 1/2"x1/2"x1/16" neodymium bar magnets, PVC glue, Gorilla Glue, hot glue, 2 3"x1' clear PVC pipes, a 6mm stainless steel ball bearing, sand paper, Dremel tool with sand paper tips, and a metal pin.

Experiment three: ABS 3D printed parts (rotor, stator, and pipe), two metal pins, a 3mm stainless steel ball bearing, 12 1"x1/2"x1/16" neodymium bar magnets, hot glue, sand paper, Dremel tool with sand paper tips, and 28 gauge copper wire.

5. Results

5a. First Experiment

The turbine rotated within its housing, but failed to remain stable and the alternator it was connected to also failed to effectively produce energy. These failings indicated problems that we iterated past in future experiments. Our alternator was changed, but we did not have the time to iterate our designs due to material and time constraints.

	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Trial 6	Average
Turbine RPM	84	114	120	106	120	92	106

Figure 1 displays the RPM of the initial turbine design under six different trials as well as the average of all six.

5b. Second Experiment

This experiment resulted in failure as the dimensions of the parts failed to remain consistent. Material tolerances were exceeded and the prints did not fit together as a result. This was a product of failed 3D prints and excessive wait times to use printers and finish post-production procedures on the SLS parts from the Fuse 1 printer. This process resulted in a failed first iteration of the stator model of the rotor, but the RPM of the system could still be found. We also used calculations in an effort to find the theoretical energy present in the system as kinetic energy. This is the energy that our turbines are aimed at harnessing and so it is the theoretical energy source for our product. We found that the pipe system would contain a total of 54.621 J available for this pipe system to harness, but we also found that the pipe system should be reduced in size to better fit homes and increase the water's velocity.

$$KE = \frac{1}{2}mv^2$$

$$m = (378\,L)(1\,kg/L) = 378\,kg$$

$$v = \frac{79.5L/min}{(\pi)(0.0381m)^2} = 0.289\,m/s$$

$$KE = \frac{1}{2}(378\,kg)(0.289\,m/s)^2$$

$$KE = 54.621\,J$$

$$\frac{P_t}{54.621\,J} = \eta$$

Figure 2 shows the equations used to calculate the kinetic energy within the pipe system in our second experiment.

	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Trial 6	Average
Turbine RPM	108	120	122	136	124	132	123 ⅔

Figure 3 displays the RPM of the second turbine design under six different trials as well as the average of all six.

5c. Third Experiment

The third iteration/experiment did not result in clear results or data, but it did prove the concept of a Straflo Turbine and demonstrated how this would effectively function. This concept was also limited by time and materials as our access to 3D printers is limited by school time and others' use of these machines. In this experiment we were again able to find the RPM of the system and able to determine the potential kinetic energy in the system. This system reduced the size of the assembly from our second iteration in an attempt to increase the water pressure of the system (figure 13) and thus the available energy. This also created problems with production and assembly however. We found that the system had access to approximately 304.51 watts. We also calculated that the system could theoretically produce at least 76 watts of power given that the system has an efficiency of at least 50%, which the EPA [8] says that most micro hydropower systems have an efficiency of 50-70%.

$$KE = \frac{1}{2}mv^2 = \frac{1}{2}(pQ\Delta t)\left(\frac{(0.408\bullet Q)}{D^2}\right)^2$$

$$= \frac{1}{2} \bullet \frac{p0.166Q^3}{D^4}\Delta t$$

$$P = \frac{(1g/cm^3)0.166(21\ g/min)^3}{2(1.5\ in)^4}$$

$$= 152\ W$$

$$P_p = 152\eta = 152(0.5)$$

$$= 76\ W$$

Figure 4 shows the equations used to find the potential wattage of the system.

	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Trial 6	Average
Turbine RPM	258	252	264	222	282	306	264

Figure 5 displays the RPM of the third turbine design under six different trials as well as the average of all six.

6. Discussion

Micro hydropower generators provide a unique opportunity to harness energy that is all around us yet remains untapped. Wastewater is a massive issue as a large portion of the water that flows through our homes everyday is unused. Harnessing the energy flowing through our pipes will make every drop of water in our homes useful. While this will not limit the amount of water that flows through our appliances, our assembly makes all of this water useful and provides an additional source of energy within the home. There have been applications of rotors within pipes to produce minimal hydroelectric power such as the Lucidpipe Power System [9]. The Lucidpipe system uses these generators in order to reduce the pressure within a system and also produce small amounts of electricity. This system mitigates the need for pressure valves as it automatically lowers the pressure while generating power. Our system similarly harnesses the power of a water system, but it does not do so in an effort to regulate pressure levels. This Lucidpipe system demonstrates alternative designs and ideas for our assembly as well as showing additional implementation and implications of our design. Other than the Lucidpipe system this idea is largely unexplored and the domestic application of micro hydropower remains untouched. These systems must be smaller so there has been no research into this area or idea that we could find. Our product and assembly can be applied in a large array of environments across the globe. This rotor can be used to harness wastewater anywhere that it flows. Whether in a home, outside the home, in commercial applications, or in agricultural applications. There are minor adjustments that may be made to fit this assembly to environments with particulates or where the wastewater may interfere with the design, but the rotor itself can be used anywhere that water is flowing. The use of this assembly is helpful in changing the wastewater that is widespread in our lives to a useful source of electricity. Following our first experiment we implemented a new form of turbine design [10] the Straflo Turbine. This new turbine design worked better as it led to greater RPM in the second and third experiment. The initial design which contained an external alternator and a system of gears had too much space for error as the gears and axles could easily be interrupted. Additionally, we found that our coils in the stator must be wound tightly around a metal core and must also be at a 1:2 ratio with the magnets. Our rotor in our third iteration was able to reach upwards of 220 RPM in 6 trials. The equations that we did for the theoretical yield of our third rotor also demonstrated that it has the capacity to produce at least 76 W and could produce more if the efficiency is maximized. Our experiments demonstrated that the Straflo Turbine configuration results in greater RPM and that the smaller pipe configuration also results in higher RPM, proving that our final experiment should provide the greatest energy production and also spin at the greatest speed while being the smallest assembly as well. This assembly is also possible to be fit to many systems as adapters can be made for greater pipe sizes in order to effectively and efficiently produce hydroelectric power.

7. Conclusions

I. In the future a vertically mounted system could not only use the power from the pressure of the pipes but also include the force of gravity. This added force on the water would increase the RPM and the overall efficiency of the system.

II. To filter out particles of solids in the future, a filtration system of some sort may need to be added to the system depending on the application. This filter would have to be accessible similar to all the other pieces of our project for easy maintenance and cleaning if necessary.

III. To increase electricity generation within our stator system, metal coil cores should be added to increase the coil conductivity. This will help with the current failures of our models. We believe hex bolts would work well so they can be screwed in and out of the system if the need arises for maintenance.

IV. When metal cores such as hex bolts are added, a drill could be used to wrap the copper wire in order to more tightly bind our wire. When the coils become more tightly bound they will be able to use more wire increasing the transfer of electrons from the spinning magnets to the copper wires.

V. Changes to the stator to allow access to the full width of the magnets as well as added room for the larger coils could also improve the electrical output of the system. Currently, the stator covers about half of the width of the magnets but with an increased width and depth of the stator cavity, our proposed larger coils could easily fit into the stator.

VI. Given more time and access, we believe that the Ender 3’s ABS 3D printing capabilities would provide the most accurate prints, resulting in the best prototypes for our system. Even after scaling the prototype down, the ABS printed pieces were not only durable but incredibly accurate. Where other versions of the project needed large amounts of sanding and other adjustments, the pieces from the Ender 3 fit together perfectly with little to no sanding.

VII. Through various changes to our system, including the switch to a stator system from the external alternator and decrease in size, the RPMs and overall efficiency of the system increased. This is due to an increase in water pressure and decrease in the number of moving parts. Additionally, if implemented into homes, the few number of parts would allow for easy maintenance should the need arise.

VIII. This project has provided sufficient evidence to the possibility of implementation of this system in homes or in a commercial setting. It was never intended to replace other forms of power, but rather harness to the water running within pipes and give it an alternative use. It is our belief that if implemented the project could provide new uses to clean and wastewater everywhere.

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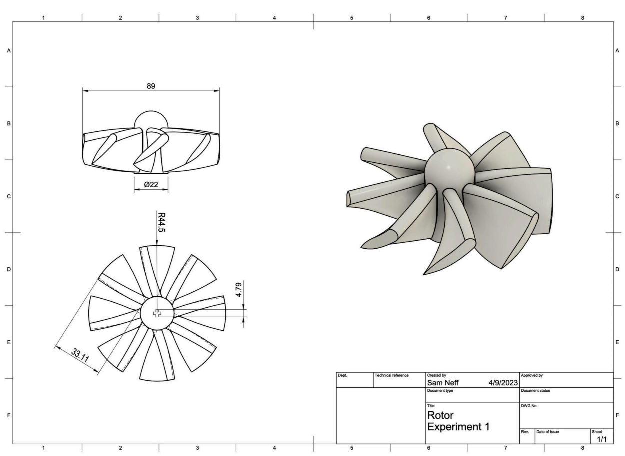
10. Annexes

Annex I. Detailed Engineering Drawings



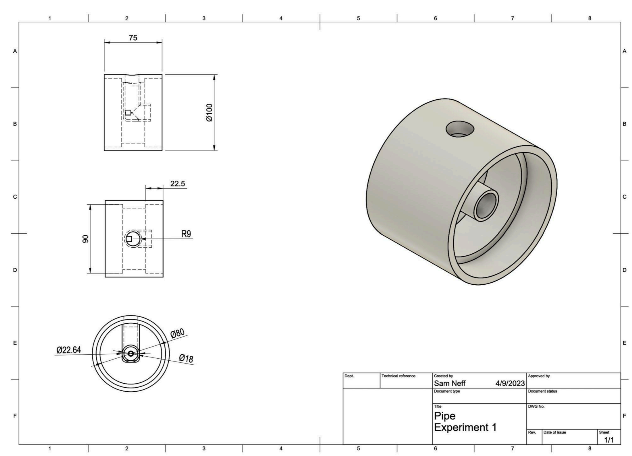
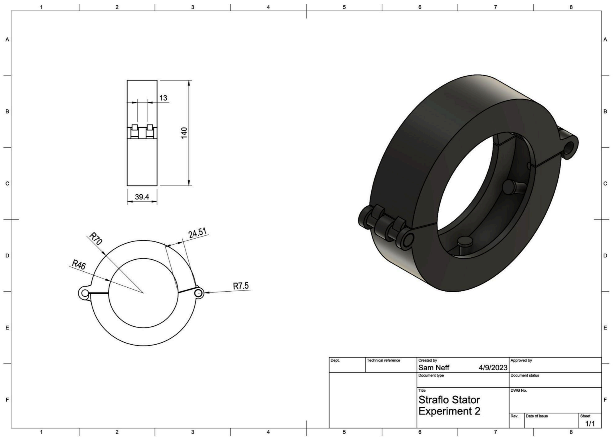


Figure 6

Figure 7



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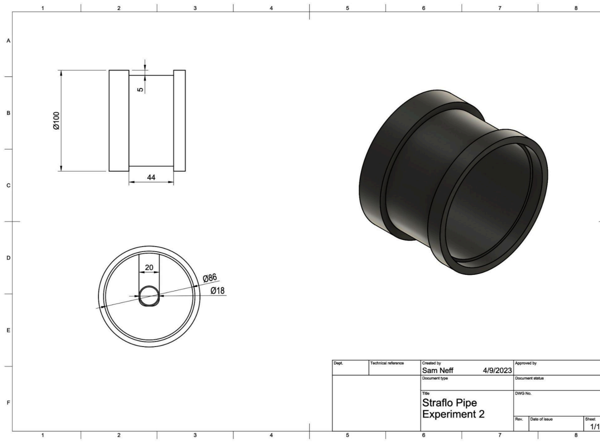


Figure 9

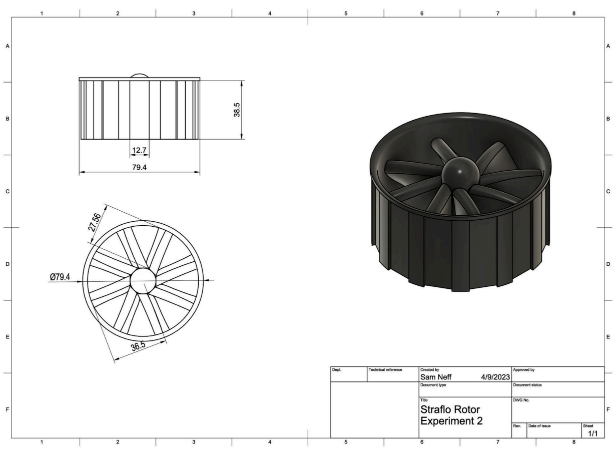


Figure 10

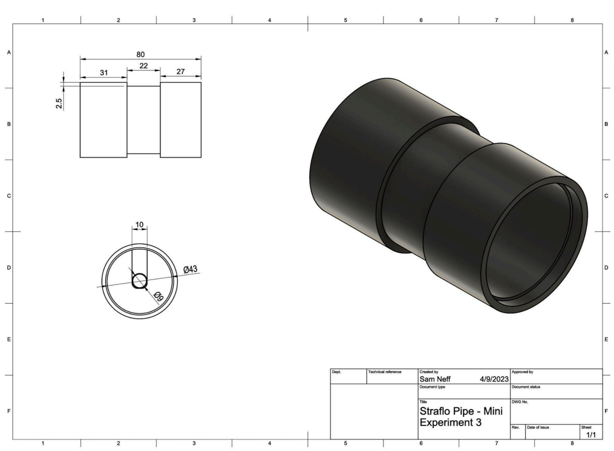
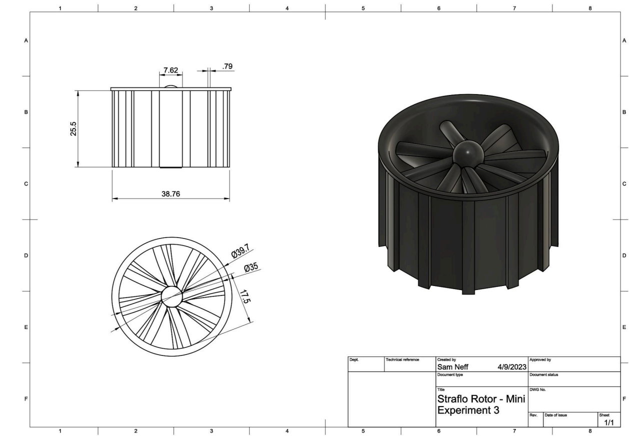


Figure 11



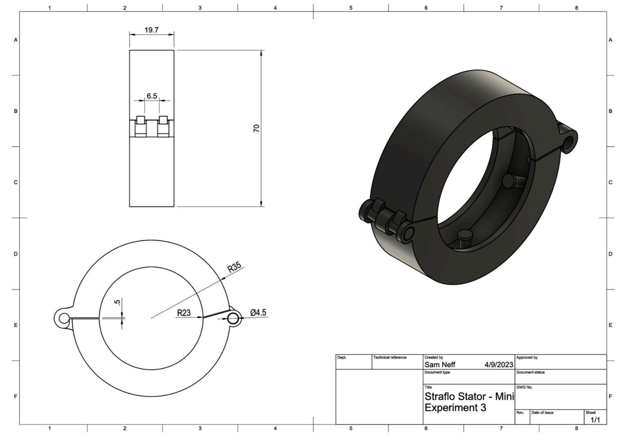
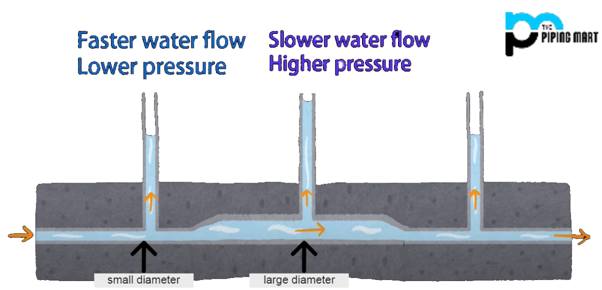


Figure 12

Figure 13



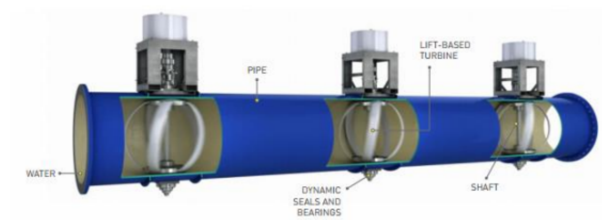


Figure 14

Figure 15